Edge-AI vs Cloud Architectures for High-Frequency Vibration Anomaly Detection in Smart Manufacturing

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ABSTRACT

Predictive maintenance in Industry 4.0 demands rapid, low-latency anomaly detection directly at the machine edge. In this paper, we benchmark an INT8 quantized 1D-CNN pipeline deployed on STM32 microcontrollers against a centralized cloud-based streaming architecture. Our edge pipeline achieves a mean inference latency of 12.4 ms with 92.1% anomaly recall on high-frequency vibration signals. However, processing continuous raw acoustic streams above 40 kHz triggers severe memory exhaustion on resource-constrained microcontrollers without downsampling. Furthermore, the reliance on single-modality 1D vibration leaves thermal expansion and optical surface wear undetected.

1. Introduction

Modern industrial automation relies increasingly on cyber-physical production systems. Unplanned downtime in automotive stamping and milling plants accounts for over $50 billion annually. While cloud-based machine learning offers virtually unlimited compute, high bandwidth costs, network jitter, and cloud outages make pure cloud architectures vulnerable for critical equipment protection.

2. System Architecture & Edge Quantization

We developed an 8-layer temporal convolutional network with dilated kernels. Using post-training INT8 quantization with representative vibration calibration sets, the model footprint was reduced from 14.2 MB to 1.1 MB. The quantized model was flashed to an ARM Cortex-M7 microcontroller operating at 480 MHz.

3. Experimental Evaluation

Benchmarking was conducted on a precision CNC spindle operating between 1,200 and 18,000 RPM. Our edge quantization pipeline delivers a 12.4 ms inference latency on STM32 microcontrollers, yielding a 92.1% anomaly recall rate while consuming only 280 mW of active power. In comparison, cloud streaming introduced an average network round-trip delay of 185 ms.

4. Limitations & Open Challenges

While our edge quantization pipeline delivers a 12.4 ms inference latency on STM32 microcontrollers, it suffers from catastrophic memory exhaustion when processing continuous raw acoustic streams above 40 kHz without severe downsampling. Furthermore, the architecture operates purely on 1D accelerometer signals and completely lacks multimodal vision fusion, rendering it incapable of detecting surface micro-cracks before thermal expansion occurs. Finally, on-device test-time model adaptation remains an open problem due to backpropagation memory limits.

5. Conclusion & Future Work

Edge-AI offers viable sub-15ms anomaly detection for rotational machinery. Future work must investigate lightweight multimodal sensor fusion and low-overhead temporal attention mechanisms capable of fitting within 512 KB of SRAM.